

Wavelet Packet Entropy as a Dynamic Biomarker for Epileptic Seizure Prediction from EEG Signals

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Abstract

Epilepsy is a chronic neurological disorder characterized by the abnormal hypersynchronization of neural activity resulting from sudden, self-sustained electrical discharges within cerebral networks. While traditional clinical approaches focus on seizure detection, a reactive approach that identifies an event as it occurs, the focus of modern research has shifted toward seizure prediction in order to allow for proactive intervention. The primary challenge lies in identifying the subtle transition from the interictal (baseline) state to the preictal (pre-seizure) state. We propose an explainable pipeline for seizure prediction using the CHB-MIT Scalp EEG Database. Our core hypothesis is that the transition to a seizure is preceded by a "complexity collapse" in brain dynamics during the preictal phase, a phenomenon that standard linear metrics often fail to detect. To quantify this effect, we use Wavelet Packet Decomposition (WPD) to achieve precise time-frequency representation of the EEG signal. We introduce Wavelet Packet Entropy (WPE) as a dynamic biomarker: whereas traditional methods search for high-energy spikes, WPE detects the increased order and reduced neural complexity characteristic of the preictal phase. The features are processed using a Random Forest classifier, chosen for its computational efficiency and its ability to provide clinically relevant insight through explainable AI (XAI) and feature importance. By employing cost-sensitive learning to mitigate the inherent class imbalance of long-term recordings, our framework is designed to support transparent and computationally efficient seizure prediction. Results suggest that WPE captures relevant pre-ictal dynamics before the onset of visible clinical symptoms, achieving a mean AUC-ROC of 0.692 across 20 evaluable patients under a Leave-One-Patient-Out evaluation framework with patient-specific calibration.

Keywords

Epilepsy, Seizure Prediction, EEG, Wavelet Packet Entropy (WPE), Random Forest, CHB-MIT,

Highlights

- Introduced a novel patient-specific polarity calibration mechanism that resolves inter-subject inconsistencies in entropy dynamics, enabling effective cross-patient seizure prediction.
- Designed a multi-component feature space integrating direction-corrected entropy, local volatility, and inter-channel spatial synchrony across 18 bipolar EEG channels.
- Evaluated the approach on the CHB-MIT Scalp EEG Database under a Leave-One-Patient-Out framework with patient-specific calibration, with aggregate ROC-based metrics reported on 20 evaluable patients.
- Achieved a mean AUROC of 0.692, suggesting that calibrated entropy biomarkers retain cross-subject predictive value in a heterogeneous pediatric cohort.
- Employed a conservative labeling protocol with a 5-minute SPH and a 4-hour postictal gap to prevent data contamination and ensure robust interictal characterization.

1. Introduction

The human brain can be described as a non-linear dynamical system, in which the coordinated activity of billions of neurons gives rise to complex electrical patterns. Electroencephalography (EEG) provides a direct, non-invasive window into these functional dynamics and remains a cornerstone of both neurological research and clinical diagnostics. Owing to its high temporal resolution and cost-effectiveness,

EEG is widely used for the long-term monitoring and management of epilepsy, a chronic neurological disorder that affects millions of individuals worldwide.

Epilepsy is characterized by recurrent, unprovoked seizures caused by sudden and abnormal hypersynchronized neuronal discharges. Its global point prevalence is estimated at approximately 0.64% of the population [1]. A substantial proportion of patients suffer from drug-resistant epilepsy, for whom conventional anti-seizure medications fail to provide adequate control. For these individuals, the unpredictable occurrence of seizures may lead to physical injury, emotional distress, and a significant reduction in quality of life. As a result, prolonged EEG monitoring is often required to support diagnosis and risk management. However, the manual inspection of long-term EEG recordings is labor-intensive, time-consuming, and inherently subjective, placing a considerable burden on clinicians and healthcare systems.

These limitations have motivated extensive research into automated EEG-based analysis systems. Most existing approaches focus on seizure detection, namely the identification of seizure events after their onset [2]. Although effective in clinical practice, such systems remain intrinsically reactive and offer limited opportunity for preventive intervention. In contrast, seizure prediction represents a more ambitious and clinically transformative objective, aiming to forecast an impending seizure by identifying the transition from the interictal (baseline) state to the preictal (pre-seizure) state. Reliable prediction would enable early warnings and timely therapeutic actions, with the potential to significantly improve patient safety and quality of life.

A central challenge in seizure prediction lies in the intrinsic properties of EEG signals. EEG recordings are highly non-stationary and non-linear, and exhibit substantial inter- and intra-subject variability, as well as contamination from physiological and environmental artifacts. Despite this complexity, many traditional approaches rely on linear or energy-based features, such as Power Spectral Density (PSD) [3]. These methods implicitly assume signal stationarity and

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primarily capture large-amplitude changes in signal power. As a result, they tend to identify seizures only after the onset of strong electrical discharges, limiting their effectiveness for early prediction [4].

To address these limitations, recent research has increasingly explored time-frequency analysis techniques, particularly wavelet-based methods such as the Discrete Wavelet Transform (DWT), for sub-band decomposition and feature extraction [3, 5]. While DWT provides improved temporal localization compared to purely frequency-domain approaches, its standard multiresolution scheme recursively decomposes only the low-frequency approximation component, potentially limiting the characterization of subtle patterns in higher-frequency bands.

In this work, we propose an alternative perspective on seizure prediction that shifts the analytical focus from signal “strength” to signal “order.” Rather than relying on large variations in amplitude or energy, we model the preictal transition as a change in the organization of EEG dynamics. To capture this phenomenon, we employ Wavelet Packet Decomposition (WPD), which extends classical wavelet analysis by recursively decomposing both approximation and detail coefficients, thus providing a more complete and uniform time-frequency representation of the EEG signal [6]. This representation enables the extraction of subtle dynamical features that may precede overt seizure activity.

Building on this representation, we use Wavelet Packet Entropy (WPE) as a dynamic biomarker of brain signal complexity [7]. Our central hypothesis is that the transition from the interictal to the preictal state is associated with a progressive neuronal hypersynchronization, leading to a measurable shift in signal complexity. While this transition is often described as a “collapse of complexity,” our analysis reveals that the direction of entropy change is not consistent across patients, highlighting a critical source of variability that is often overlooked. To address this issue, a central contribution of this work is the introduction of a patient-specific polarity calibration mechanism, which explicitly models and corrects inter-subject differences in entropy dynamics, enabling consistent cross-patient learning.

Furthermore, we design a multi-component feature representation that extends beyond instantaneous entropy by incorporating its temporal evolution, local variability, and spatial synchrony across EEG channels. This structured feature space captures complementary aspects of the preictal transition, allowing the model to represent not only where changes occur, but also how they evolve over time and how they propagate across brain regions. For classification, we adopt a Random Forest model, which offers a favorable trade-off between predictive performance, computational efficiency, and interpretability [8, 9]. In clinical settings, interpretability is a key requirement, as it enables clinicians to relate model outputs to underlying physiological mechanisms. In this regard, the use of feature importance analysis provides additional insight into the spatial and dynamical characteristics of epileptogenic activity.

The proposed framework is evaluated on the CHB-MIT Scalp EEG Database, a widely used benchmark dataset containing long-term EEG recordings from pediatric patients with epilepsy. The evaluation is conducted under a Leave-One-Patient-Out protocol with subject-level calibration, providing a realistic assessment of cross-subject generalization. The results indicate that entropy-based features, when properly aligned through polarity calibration, retain predictive value across heterogeneous patients while preserving inter-

pretability. In this way, the proposed approach contributes not only to seizure prediction, but also to the understanding of the dynamical processes underlying seizure generation, supporting the development of clinically applicable and transparent decision-support systems.

2. Related Works

2.1. Seizure Detection and Seizure Prediction

Automated EEG-based seizure analysis has been investigated mainly in two related but distinct tasks: seizure detection and seizure prediction [4, 10]. Although these tasks are sometimes discussed together, they address different clinical objectives. Seizure detection aims to identify ictal activity after its onset, whereas seizure prediction seeks to recognize the transition from the interictal state to a preictal condition before seizure symptoms appear. This distinction is important since prediction relies on subtle dynamic changes, whereas detection focuses on large amplitude variations typically associated with seizure onset.

2.2. Machine Learning and Deep Learning Approaches

Early automated EEG analysis systems were largely based on classical machine learning (ML) pipelines that combined features with conventional classifiers [2, 5]. Spectral and nonlinear descriptors, extracted from short EEG epochs, were used with classifiers such as k-Nearest Neighbors or Random Forests [11]. These approaches remain attractive because they are computationally efficient and often provide features that are easier to interpret. However, their performance depends on feature engineering choices and may decline when applied to heterogeneous cross-subject datasets [12].

More recently, deep learning (DL) models have been widely adopted for this task. DL or hybrid architectures - such as Convolutional Neural Networks and Long Short-Term Memory networks - have shown strong performance in both settings, mainly because they learn hierarchical representations directly from raw EEG or minimally processed inputs, reducing the need for manual feature engineering while capturing complex patterns in the data. However, as noted by Kerr et al. [4], they generally require larger datasets and higher computational costs, and usually provide limited interpretability regarding which signal characteristics drive the final decision.

2.3. Time-Frequency and Entropy-Based Features

Since EEG signals are intrinsically non-stationary, time-frequency methods have become a central component of seizure analysis. Wavelet-based techniques are particularly well suited to this setting, as they provide localized information in both time and frequency. In many previous studies, DWT is often the starting point for refining the EEG signal: it allows the relevant components to be isolated from background noise, breaking down brain activity into sub-bands that highlight otherwise invisible details [5, 6]. However, the standard DWT recursively decomposes only the approximation branch, which may limit its ability to capture transient

patterns in higher-frequency components. In addition to the approximation coefficients considered by DWT, WPD also decomposes the detail coefficients, thereby providing a more complete and uniform representation of EEG dynamics across frequency bands [7]. This richer decomposition is relevant in seizure-related applications, where informative changes may occur across multiple spectral regions rather than being confined to low-frequency components alone.

Alongside wavelet analysis, entropy-based measures have also been used to quantify the complexity of EEG activity and its relative dynamics. Among these, Wavelet Entropy and related nonlinear descriptors are particularly useful because they capture not only signal power, but also how energy is distributed across frequency sub-bands. In the context of epilepsy, this aspect is particularly relevant, as the transition to seizure onset may be associated with a neuronal hypersynchronization and a reduction in signal complexity [10, 4]. For this reason, entropy-based features may therefore help detect preictal changes that may not be visible through purely linear or energy-based measures [11].

2.4. Interpretability and Validation Protocols

Beyond feature extraction and model architecture, two methodological aspects are especially important in automated EEG analysis: interpretability and validation strategy. In clinical applications, models should ideally provide not only accurate predictions, but also outputs that can be related to clinically relevant signal characteristics [4]. For this reason, transparent classifiers remain relevant, as they combine robustness with the possibility of feature-level importance analysis [13].

A second methodological issue concerns how model performance is evaluated. When recordings from the same patient are included in both the training and test sets, performance may be overestimated because the model can partly exploit subject-specific patterns instead of learning seizure-related dynamics that generalize across individuals. This is particularly relevant in EEG analysis, which shows strong inter-subject variability in baseline activity. As a result, cross-subject protocols such as Leave-One-Patient-Out (LOPO) generally provide a more conservative and clinically realistic estimate of generalization performance [12, 4].

2.5. Research Gap and Motivation of This Work

The literature reviewed above highlights several open challenges in EEG-based seizure prediction research. Many studies still focus on seizure detection rather than prediction, thereby addressing a more reactive clinical objective [2, 4]. More generally, although seizure prediction has long been recognized as a clinically important goal, identifying reliable preictal markers remains methodologically challenging [10]. In addition, some studies rely on validation settings that do not fully test generalization to unseen patients [12]. Furthermore, although DL methods have become increasingly prominent, they are less interpretable, whereas classical feature-based methods may not fully capture the non-stationary and nonlinear nature of EEG signals [5].

To address these limitations, this work introduces an entropy-based seizure prediction framework that explicitly

models preictal dynamics through WPE and incorporates a novel patient-specific polarity calibration mechanism to enable robust cross-subject generalization while preserving interpretability.

3. Methodology

3.1. Pipeline Overview

The proposed framework analyzes multi-channel EEG recordings through four sequential steps, from raw signal acquisition and ending with binary preictal/interictal epochs classification. The pipeline combines signal preprocessing, time-frequency decomposition, nonlinear complexity quantification, and supervised classification, to develop an interpretable and reproducible system for seizure prediction.

Given a multi-channel EEG recording $X(t) = \{x_1(t), x_2(t), \dots, x_C(t)\}$, the proposed pipeline operates on temporally segmented epochs. The underlying assumption is that, although EEG signals are globally non-stationary, they can be treated as locally stationary within sufficiently short time windows [14]. This allows the extraction of stable spectral features from each epoch independently, without requiring the signal to be stationary over the entire recording. Each epoch is then transformed into a compact feature representation that captures frequency-dependent complexity variations across channels.

The pipeline consists of four main stages:

1. **Temporal segmentation and preprocessing:** Continuous EEG recordings are first filtered and partitioned into overlapping windows. At this stage artifact attenuation is applied to reduce the influence of non-neural noise sources, yielding segments that are suitable for subsequent time-frequency analysis.
2. **Wavelet Packet Decomposition (WPD):** Each EEG segment is decomposed to a full depth of levels using the Daubechies-4 (db4) mother wavelet, providing a uniform tiling of the time-frequency plane [15, 11]. To ensure that the resulting feature vector represents a sequential increase in frequency bands (e.g., 0-4 Hz, 4-8 Hz, etc.), we implement a frequency-based node ordering (also known as Paley or Gray ordering) using the `order='freq'` parameter in the PyWavelets library. To minimize edge artifacts during decomposition, we employ symmetric boundary handling (signal extension), reducing spectral leakage at segment boundaries.
3. **Complexity quantification via Wavelet Packet Entropy (WPE):** For each channel and epoch, the energies of the terminal sub-bands are normalized to form a probability distribution, and Shannon entropy is computed over it using the natural logarithm. To ensure numerical stability, a small regularization constant, equal to $1e-10$, is added to handle zero-energy sub-bands. The resulting scalar, expressed in nats and not normalized to the range, indicates how broadly or narrowly the signal energy is distributed across frequency sub-bands, thus providing a measure of spectral complexity. In our study, this entropy value is the main biomarker under investigation [16, 11].
4. **Classification and explainability:** Channel-wise entropy values are concatenated into a feature vector, which is passed to a Random Forest classifier

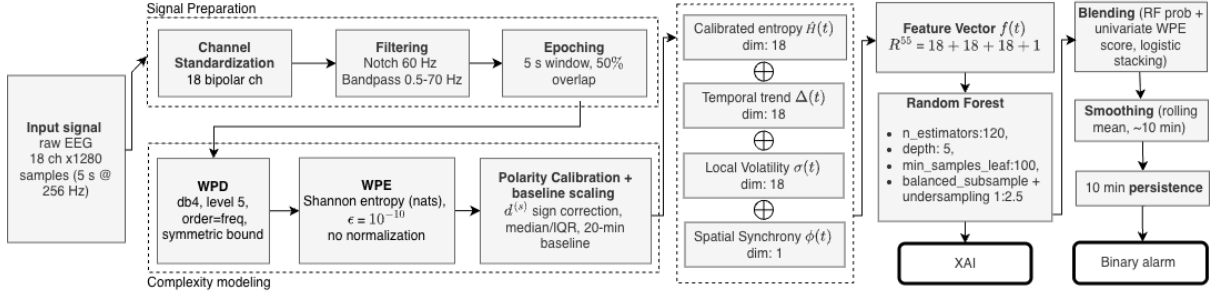


Figure 1: Proposed Pipeline Overview, illustrating the flow from raw EEG acquisition to seizure prediction and clinical insight via feature importance analysis

for binary classification between interictal and preictal states. Beyond its classification performance, the Random Forest provides feature importance estimates, allowing to identify the most informative channels and relate them to specific scalp regions. This adds a layer of transparency where model decisions need to be interpretable by practitioners [13].

The central hypothesis underlying the pipeline is that the transition toward seizure onset is marked by a progressive reduction in spectral complexity, reflected in a measurable decrease in WPE. As the brain dynamics shift from the interictal to the preictal state, neural activity is assumed to become increasingly synchronised and more narrowly distributed across frequency bands, leading to lower entropy values. The primary objective of the system is to detect this entropy drop before the emergence of clinical symptoms. The modular structure of the pipeline further strengthens its practical value, since each processing block can be evaluated and adjusted independently, allowing the framework to be adapted to different datasets or electrode configurations without redesigning the full analysis chain.

3.2. Dataset

The dataset employed in this study is the CHB-MIT Scalp EEG Database [17, 18], a publicly available benchmark dataset hosted on PhysioNet and widely used in epilepsy research for seizure detection and prediction [12, 19]. The recordings were collected at Boston Children’s Hospital from pediatric patients diagnosed with epilepsy, many of whom present drug-resistant forms of the disorder.

As reported in Table 1, the dataset comprises EEG recordings from 24 cases corresponding to 23 individual patients (5 males aged 3-22 years and 17 females aged 1.5-19 years). Case chb21 was obtained 1.5 years after case chb01, from the same female subject [20]. EEG signals were recorded using the international 10-20 electrode placement system with scalp electrodes, sampled at 256 Hz and stored with a 16-bit resolution. Each recording session lasts between 1 and 4 hours, resulting in over 900 hours of continuous EEG signals data.

A total of 198 seizure events are annotated in the dataset, with seizure durations ranging from a few seconds to several minutes. The seizure types include focal and generalized seizures, originating from different cortical regions, that cause substantial variability in spatial and temporal seizure characteristics. The dataset also includes a large number of non-seizure recordings, reflecting the predominance of interictal activity in long-term EEG monitoring.

Table 1

Summary of the CHB-MIT Dataset Characteristics

Variable	Value
Overview	
Total Subjects	23
Total EEG Cases	24
Total Seizure Events	198
Total Recording Duration	> 900 hours
Gender*	
Female n(%)	17 (77.27%)
Male n(%)	5 (22.72%)
Missing n	1
Age (years)*	
Mean(sd)	9.98(5.55)
Median(q1-q3)	10 (6-14)
Range(min-max)	1.5-22
Missing(n)	1
Seizure Types	Focal, Generalized

* Demographic information for Subject 24 is not provided in the original database metadata.

^a Subjects 01 and 21 refer to the same patient, recorded at a distance of 1.5 years.

EEG recordings were acquired using variable channel configurations, with the number of channels per recording ranging from approximately 18 to 26, depending on the acquisition setup. In some recordings, additional non-EEG signals such as electrocardiography (ECG) or stimulation channels are present alongside EEG signals. Only scalp EEG channels are considered in this study. As mentioned before, a defining characteristic of the CHB-MIT dataset is its pronounced class imbalance: seizure activity accounts for less than 1% of the total recording duration. While this imbalance poses a significant challenge for automated analysis, it reflects realworld clinical conditions and underscores the importance of developing a robust and reliable seizure prediction system capable of operating on long-term continuous EEG data.

3.3. Preprocessing

EEG preprocessing was conducted to ensure consistency across recordings while preserving the intrinsic dynamical properties of the signal. All preprocessing procedures were implemented in Python using the MNE and PyWavelets libraries.

3.3.1. Channel Standardization and Cleaning

The CHB-MIT database shows a marked variability in channel configurations across subjects and recording sessions. To ensure spatial consistency throughout the analysis, only scalp EEG channels were considered. Therefore the non-EEG signals such as ECG, VNS, EKG, and status indicators were excluded from all recordings. A standardized bipolar montage of 18 channels was adopted, following configurations commonly used in the literature for this dataset [21, 22]: FP1-F7, F7-T7, T7-P7, P7-O1, FP1-F3, F3-C3, C3-P3, P3-O1, FP2-F4, F4-C4, C4-P4, P4-O2, FP2-F8, F8-T8, T8-P8, P8-O2, FZ-CZ, CZ-PZ. The bipolar reference was chosen because it enhances spatial specificity by computing potential differences between adjacent electrodes, which reduces common-mode artifacts and reference-related contamination. This is relevant for seizure prediction, where the goal is to capture localized cortical dynamics rather than global voltage fluctuations referenced to a single electrode [5]. Some recordings contain naming inconsistencies across sessions, for instance the T8-P8-1 variant appearing in place of T8-P8. A channel-matching procedure was implemented to identify and remap such aliases to the standard configuration. Recordings that could not be completed to the full 18-channel set after remapping were excluded from the analysis. This strict standardization was necessary to maintain a fixed-dimensional feature space across all subjects, avoiding mismatches during classifier training [2].

3.3.2. Filtering and Noise Suppression

Signal filtering followed a two-stage procedure consistent with standard seizure analysis pipelines [2]. A notch filter was applied at 60 Hz to remove power-line interference, a common source of spectral contamination in clinical EEG recordings that would otherwise distort the time-frequency analysis, and the signal was then bandpass filtered between 0.5 Hz and 70 Hz. The lower cutoff removes slow baseline drifts and DC offsets, while the upper cutoff attenuates high-frequency muscular artifacts. This range covers the full spectrum of clinically relevant neural oscillations, from delta to gamma, and preserves potential high-frequency epileptiform components that may carry preictal information [5]. No further artifact rejection or aggressive denoising was applied. This was a deliberate choice: overly aggressive preprocessing can distort the nonlinear structure of the signal, which is precisely what entropy-based measures are designed to capture [14]. In this context, preserving the complexity of the raw dynamics takes priority over obtaining a morphologically clean trace.

3.3.3. Epoching Strategy and Labeling

After filtering, continuous recordings were segmented into fixed-length epochs of 5 seconds. This window length is within the range commonly adopted in seizure prediction frameworks, as it provides a reasonable trade-off between temporal resolution and the statistical stability of the extracted features [23]. Consecutive windows were overlapped by 2.5 seconds, corresponding to a 50% overlap. Sliding-window approaches with partial overlap allow smoother temporal tracking of dynamical transitions, reducing sudden discontinuities in the entropy time series while keeping computational cost manageable. At a sampling frequency of 256 Hz, each epoch contains 1280 samples per channel. Segmenting the signal into fixed-length epochs

guarantees a consistent feature-space dimension across all samples, which is required for subsequent ML analysis [2].

3.3.4. Temporal Labeling and Exclusion Criteria

Epoch labeling was performed with respect to the clinically annotated seizure onset times, provided in the CHB-MIT metadata. To avoid temporal contamination between classes, explicit exclusion zones were defined around each seizure event. A Seizure Prediction Horizon (SPH) of 5 minutes immediately preceding each annotated seizure onset was excluded from the analysis. Epochs falling within this interval are neither clearly interictal nor fully preictal, and including them would introduce ambiguity at the class boundary. The preictal label was therefore assigned to the 30-minute window preceding the SPH, consistently with labeling conventions adopted in comparable studies. A postictal exclusion gap of 4 hours was applied after each seizure. EEG activity in the postictal phase is known to deviate substantially from the interictal baseline due to neuronal refractoriness and residual synchronization effects. Including postictal epochs as interictal samples would introduce systematic bias into the baseline class, blurring the distinction the classifier is trained to make. In recordings with closely spaced seizures, temporal overlap between preictal and postictal exclusion zones was resolved by removing the affected epochs entirely. This conservative strategy prioritises label purity over data quantity, which is preferable given the already severe class imbalance characteristic of long-term EEG recordings.

3.3.5. Wavelet Packet Entropy as a Unified Dynamical Biomarker

Let $x_c^{(n)}(t)$ denote the EEG signal acquired from channel $c \in \{1, \dots, 18\}$ within epoch n , where each epoch has fixed duration T seconds and sampling frequency $f_s = 256$ Hz, giving $N = T \cdot f_s$ samples per epoch.

WPD was applied to $x_c^{(n)}(t)$ using the Daubechies 4 (db4) mother wavelet at decomposition level $L = 5$. WPD recursively decomposes both approximation and detail coefficients, producing a complete binary tree with $2^L = 32$ terminal sub-bands at level L . This uniform frequency partition covers the full bandwidth of the filtered signal and has been shown to provide better time-frequency resolution than standard DWT for epileptic EEG analysis [23, 10].

Let $w_{i,k}^{(c,n)}$ denote the k -th wavelet packet coefficient at node i of level L , for channel c and epoch n . The energy of the i -th sub-band is:

$$E_i^{(c,n)} = \sum_k \left| w_{i,k}^{(c,n)} \right|^2 \quad (1)$$

The total energy across all terminal nodes is:

$$E_{tot}^{(c,n)} = \sum_{i=1}^{2^L} E_i^{(c,n)} \quad (2)$$

Normalizing the sub-band energies yields a probability mass function over the 2^L nodes:

$$p_i^{(c,n)} = \frac{E_i^{(c,n)}}{E_{tot}^{(c,n)}} \quad (3)$$

Wavelet Packet Entropy (WPE) is then computed as the Shannon entropy of this distribution:

$$H^{(c,n)} = - \sum_{i=1}^{2^L} p_i^{(c,n)} \ln(p_i^{(c,n)}) \quad (4)$$

The quantity $H^{(c,n)}$ measures how uniformly signal energy is distributed across frequency sub-bands. The entropy is expressed in nats, as the natural logarithm is used for computation. We chose not to normalize the final entropy values to a fixed range. This decision preserves the absolute scale of the signal's dynamical complexity, which is essential for capturing the "complexity collapse" phenomenon across different subjects. These raw values are subsequently handled by our patient-specific robust scaling during the feature engineering phase (see subsection 3.4.1).

To maintain robustness across all recording conditions, the probability mass function incorporates a small constant ($1e-10$) to prevent undefined logarithms in nodes with zero energy, ensuring the pipeline remains numerically stable without significantly biasing the entropy estimates. When spectral energy is spread broadly entropy is high, reflecting a complex and diverse dynamical regime. When energy concentrates in a few sub-bands, as occurs during synchronised rhythmic discharges, entropy decreases, reflecting increased regularity. This relationship between hypersynchronisation and entropy reduction has been reported in both scalp and intracranial EEG studies [21, 22].

Computing $H^{(c,n)}$ independently for each of the 18 channels yields a feature vector per epoch:

$$\mathbf{H}^{(n)} = [H^{(1,n)}, H^{(2,n)}, \dots, H^{(18,n)}] \quad (5)$$

Stacking all epochs produces a feature matrix of dimensions $N_{epochs} \times 18$, which serves as the input to the subsequent feature engineering stage. This entropy-based representation provides a compact dynamical summary of each epoch, sensitive to the spectral reorganisation that characterises the preictal transition [14].

3.4. Feature Engineering and Polarity Calibration

Starting from the raw 18-dimensional WPE vector computed for each epoch, a structured feature engineering procedure was applied to enrich the temporal and spatial expressiveness of the representation. A central issue addressed at this stage is the inter-patient variability in the direction of the preictal entropy shift, which requires explicit handling before cross-patient classification can be attempted.

3.4.1. Patient-Specific Polarity Calibration

One issue we did not anticipate initially is that patients differ not just in how much WPE changes before a seizure, but in which direction. While the complexity collapse hypothesis suggests that entropy should drop before a seizure, our data showed that it isn't a universal rule. In a subset of patients we observed the exact opposite: WPE values actually trended higher during the preictal window, systematically diverging from the interictal baseline. This inter-individual difference is not an artifact. In fact, it likely reflects genuine heterogeneity in how different patients' brains reorganise prior to a seizure. However, if left uncorrected, it causes the cross-patient classifier to receive contradictory training signals: preictal epochs from patients with opposite polarities partially cancel each other out in feature space, degrading generalisation.

To address this, a patient-specific polarity calibration step was introduced. For each subject s , the mean WPE was computed separately over all preictal epochs $\mu_{pre}^{(s)}$ and over

all interictal epochs $\mu_{inter}^{(s)}$, using the subject's complete recording:

$$\delta^{(s)} = \mu_{pre}^{(s)} - \mu_{inter}^{(s)} \quad (6)$$

A binary direction indicator was then assigned based on the sign of $\delta^{(s)}$:

$$d^{(s)} = \begin{cases} +1 & \text{if } \delta^{(s)} > 0 \\ -1 & \text{if } \delta^{(s)} \leq 0 \end{cases} \quad (7)$$

Each subject's WPE matrix was multiplied element-wise by $d^{(s)}$, producing a direction-corrected representation $\hat{\mathbf{H}}^{(s)}$ in which the preictal transition is encoded as a positive deviation from the interictal baseline, regardless of the subject's native entropy polarity. The relative temporal structure of the entropy signal is preserved, only the sign is flipped where necessary.

To further stabilise the distribution of the corrected values and reduce the influence of outliers, a patient-specific robust scaling was applied. To prevent data leakage and ensure a causally valid pipeline, the normalization parameters (specifically the median and the interquartile range (IQR)) were fitted independently for each patient using only an initial 20-minute baseline segment of interictal data. This "frozen" calibration was then applied to the entire recording. The resulting representation, denoted $\hat{\mathbf{H}}^{(s)} \in \mathbb{R}^{N_{epochs} \times 18}$, constitutes the base feature block used in all subsequent steps.

Clinically, this step mirrors an initial calibration session, in which a short labeled EEG segment (containing at least one seizure) is used to establish a patient's characteristic entropy direction before the monitoring system is deployed. This is broadly consistent with how automated prediction tools are typically activated only after an initial observation period in clinical practice. We therefore treat the proposed framework as a shared classifier operating after a subject-level enrollment step, not as a fully blind, zero-calibration predictor.

To the best of our knowledge, explicitly modeling and correcting inter-patient variability in the direction of entropy change through a polarity calibration step has not been systematically addressed in previous seizure prediction frameworks. This component plays a crucial role in enabling the use of entropy-based features in a cross-subject setting.

3.4.2. Temporal Trend Features

The calibrated WPE values capture the instantaneous spectral complexity of each epoch, but they do not directly encode how entropy is evolving over time relative to the patient's long-term baseline. To capture this temporal dynamic, a dual-timescale moving average difference was computed on the direction-corrected, quantile-normalised matrix $\hat{\mathbf{H}}^{(s)}$.

Let $\hat{h}_c(t)$ denote the sequence of corrected WPE values for channel c over time. Two moving averages were defined over different temporal scales:

$$MA_{fast}(t) = \frac{1}{W_f} \sum_{\tau=0}^{W_f-1} \hat{h}_c(t - \tau), \quad W_f = 12 \quad (8)$$

$$MA_{slow}(t) = \frac{1}{W_s} \sum_{\tau=0}^{W_s-1} \hat{h}_c(t - \tau), \quad W_s = 480 \quad (9)$$

The fast window $W_f = 12$ epochs corresponds to 30 seconds of EEG (given a step size of 2.5 seconds per epoch) and captures short-term fluctuations in entropy. The slow window $W_s = 480$ epochs spans approximately 20 minutes and provides a stable estimate of the subject's inter-ictal entropy baseline. The temporal trend feature is then defined as the signed difference between the two:

$$\Delta_c(t) = MA_{fast}(t) - MA_{slow}(t) \quad (10)$$

A positive $\Delta_c(t)$ indicates that local entropy has risen above the long-term baseline, while a negative value reflects a transient suppression. After polarity calibration, an increasing $\Delta_c(t)$ consistently signals proximity to seizure onset across all subjects. This block adds an additional 18-dimensional component to the feature representation.

3.4.3. Local Volatility

To complement the directional trend, a local volatility feature was introduced to capture the short-term instability of the entropy signal. For each channel c , the rolling standard deviation over the fast window was computed:

$$\sigma_c(t) = \sqrt{\frac{1}{W_f - 1} \sum_{\tau=0}^{W_f-1} \left(\hat{h}_c(t - \tau) - MA_{fast}(t) \right)^2} \quad (11)$$

Elevated values of σ may reflect increased short-term instability in the entropy signal during the transition toward seizure onset [10, 4]. Unlike the trend feature, $\sigma_c(t)$ is sign-invariant and therefore independent of the polarity calibration, providing a complementary view of preictal dynamics that does not rely on the direction of the entropy shift. This yields a further 18-dimensional feature block.

3.4.4. Spatial Synchrony

The three feature blocks described above treat each channel independently. The spatial synchrony feature instead captures the degree to which all 18 channels simultaneously exhibit similar entropy values, providing a measure of global network coherence. Progressive homogenisation of entropy across channels is consistent with the large-scale changes in network synchronization often reported before seizure onset [24, 10].

For each epoch t , the instantaneous standard deviation across channels was computed:

$$\phi(t) = \text{std}_c(\hat{\mathbf{H}}^{(s)}(t)) = \sqrt{\frac{1}{18} \sum_{c=1}^{18} \left(\hat{h}_c(t) - \bar{h}(t) \right)^2} \quad (12)$$

where $\bar{h}(t)$ is the mean WPE across all 18 channels at epoch t . A high value of $\phi(t)$ indicates spatially heterogeneous entropy: different brain regions are operating in different dynamical regimes, while a low value suggests convergence toward a uniform entropy level across the scalp, which is consistent with global synchronisation. To reduce the effect of transient fluctuations, a rolling mean of $\phi(t)$ over the fast window W_f was applied, yielding a single scalar feature per epoch.

3.4.5. Final Feature Representation

The complete feature vector for each epoch is obtained by concatenating all four components:

$$\mathbf{f}(t) = [\hat{\mathbf{H}}^{(s)}(t), \delta(t), \sigma(t), \phi(t)] \in \mathbb{R}^{55} \quad (13)$$

where $\hat{\mathbf{H}}^{(s)}(t) \in \mathbb{R}^{18}$ are the direction-corrected quantile-normalised WPE values, $\Delta(t) \in \mathbb{R}^{18}$ are the temporal trend features, $\sigma(t) \in \mathbb{R}^{18}$ are the local volatility features, and $\phi(t) \in \mathbb{R}^1$ is the spatial synchrony scalar. The full feature matrix over all epochs has dimensions $N_{epochs} \times 55$.

Each component encodes a distinct aspect of the preictal dynamics: the calibrated WPE captures the direction of entropy change, the temporal trend quantifies its deviation from the long-term baseline, the volatility measures its local instability, and the spatial synchrony reflects the degree of inter-channel coherence. Together, these four descriptors provide the classifier with a feature space that covers both the temporal evolution and the spatial organisation of the preictal transition.

3.5. Classification Framework and Model Validation

The feature matrix described in section 3.4 was used as input to a supervised classification framework designed to discriminate between interictal and preictal states. This section describes the normalization strategy, the approach adopted to handle class imbalance, the classifier configuration and the validation protocol.

The operating threshold used to binarize predictions was selected using Youden's J statistic, computed via a nested cross-validation performed exclusively on training-patient data within each LOPO fold. This ensures that no information from the held-out test patient influences threshold selection, and that the reported sensitivity, specificity, and FAR/h estimates are not tuned on the evaluation segment itself.

3.5.1. Feature Normalization

Feature normalization was handled within the patient-specific construction pipeline. As described in subsection 3.4.1, after polarity correction, each subject's 18-channel WPE matrix was standardized using robust scaling. The parameters (median and IQR) were calculated exclusively from the first 20 minutes of the recording's interictal data, and the resulting transformation was fitted independently for each patient and then applied to that patient's entire direction-corrected WPE matrix. This choice was made to reduce the effect of heavy-tailed values and inter-subject scale differences, which are common in long-term EEG recordings, while strictly preventing data leakage by ensuring that no future distributional information from the evaluation segments influenced the scaling process. To further mitigate the impact of extreme outliers, all features were clipped to the $[-5, 5]$ range after scaling. This approach effectively simulates a real-time clinical scenario where calibration parameters are fixed at the start of the monitoring period.

The resulting normalized matrix was used as the base feature block for the temporal trend, local volatility, and spatial synchrony features described in subsections 3.4.2, 3.4.3 and 3.4.4. Because the normalization was estimated independently for each patient after polarity correction, it did not transfer feature statistics across subjects. It should therefore be interpreted as a subject-specific calibration step rather than as a fully blind, zero-calibration setting.

3.5.2. Class Imbalance Handling

A well-known challenge in long-term EEG analysis is the severe imbalance between seizure-related and baseline epochs. As mentioned above, in the CHB-MIT dataset seizure activity accounts for less than 1% of the total recording duration, which leads to a heavily skewed class distribution. A classifier trained without accounting for this imbalance will tend to predict the majority class for almost all epochs, achieving high accuracy while being essentially useless for seizure prediction.

To address this, a hybrid strategy combining cost-sensitive learning and controlled undersampling was adopted. Class weights were assigned inversely proportional to class frequencies:

$$w_k = \frac{N}{K \cdot N_k} \quad (14)$$

where N is the total number of training samples, K the number of classes, and N_k the number of samples in class k . This weighting increases the penalty for misclassifying preictal epochs during training, pushing the decision boundary toward better minority class coverage.

In addition to cost-sensitive weighting, a structured undersampling procedure was applied to further reduce the dominance of the interictal class. All available preictal epochs were retained, while interictal epochs were randomly subsampled to maintain a fixed 1:2.5 ratio between preictal and interictal samples in each training set. This ratio was chosen to prevent the model from being overwhelmed by the baseline distribution while still exposing it to sufficient variability within the interictal class. The combination of these two strategies provides a controlled trade-off between sensitivity to the minority class and stability of the decision boundary.

3.5.3. Classifier Architecture and Hyperparameters

We selected Random Forest mainly because it is robust to noisy and correlated predictors, can control overfitting through depth and leaf-size constraints, and provides native feature-importance estimates that support post-hoc interpretability [13]. The ensemble is composed of decision trees trained via bootstrap aggregation, where each tree is built from a random subset of features and training samples. Predictions are obtained by majority voting across all trees, and class posterior probabilities are estimated as the fraction of trees voting for each class.

The hyperparameter configuration was fixed across all LOPO folds to ensure comparability of per-patient results and is summarised in Table 2.

Table 2
Random Forest hyperparameter configuration.

Parameter	Value	Rationale
n_estimators	120	Stable ensemble estimates.
max_depth	5	Prevents overfitting.
min_samples_leaf	100	For stable probability estimates.
class_weight	balanced_subsample	Per-tree rebalancing.
random_state	42	Ensures reproducibility of results.

The max_depth constraint of 5 levels plays a particularly important role in the cross-patient setting. By limiting tree depth the model is prevented from memorising patient-specific EEG patterns present in the training data, which encourages learning of more generalisable preictal

signatures. Similarly, min_samples_leaf = 100 ensures that each terminal node is supported by at least 100 training epochs, providing statistically stable probability estimates even in the presence of noisy or ambiguous preictal segments. The balanced_subsample class weighting strategy differs from standard balanced weighting in that class frequencies are recomputed independently within each bootstrap sample rather than globally over the full training set. With balanced_subsample, class weights are recomputed within each bootstrap sample, reducing the risk that subjects with much longer interictal recordings dominate the effective training distribution. To further stabilize predictions, the Random Forest probability was combined with a univariate score derived from the mean calibrated WPE via a secondary logistic regression (stacking), fitted exclusively on training-patient data. The final probability estimate was computed as a weighted blend of the two components ($\alpha = 0.7$ for the Random Forest term), prior to temporal smoothing.

3.5.4. Cross-Subject Validation Protocol

To assess how well the proposed framework generalises to unseen patients, a LOPO Cross-Validation (LOPO-CV) protocol was employed. At each fold, given n the number of samples, the complete recording of one subject is held out exclusively for testing, while the remaining $n - 1$ subjects form the training set. This is repeated for all the subjects, so that every patient serves as the test case exactly once.

LOPO-CV is the most conservative form of cross-validation for clinical EEG studies, as it explicitly prohibits any information from the test subject from influencing model training [12, 4]. This is particularly relevant in seizure prediction, where intra-subject correlations in EEG morphology can lead to overly optimistic performance estimates if the same subject contributes to both training and evaluation.

All patient-specific preprocessing steps, including polarity calibration, robust scaling and temporal feature computation, were applied independently per patient prior to the LOPO split. The polarity direction $d^{(s)}$ and the scaling parameters for each subject are fitted exclusively on that subject’s own data and do not incorporate any information from other patients. $d^{(s)}$ is estimated from each patient’s complete recording, including epochs that later appear in the test set. This is within-patient information usage, not cross-patient leakage, indeed it does not affect the shared Random Forest’s parameters or structure. This is consistent with a clinical enrollment scenario in which a brief labeled baseline is used to configure the monitoring system for each individual patient before deployment; we further discuss the implications of this design choice for real-time causality in section 6.

To reduce the effect of epoch-level noise in the classifier output, a temporal smoothing procedure was applied to the predicted probability sequence of each test subject. Specifically, a centered rolling mean with a window of 240 epochs (approximately 10 minutes) was computed over the raw per-epoch probabilities prior to AUC evaluation. This smoothing attenuates high-frequency fluctuations in the output and better reflects the gradual nature of preictal transitions. Additionally, the first 480 epochs (approximately 20 minutes) of each test recording were excluded from evaluation to allow the slow moving average baseline MA_{slow} to stabilise before assessment begins.

Although the LOPO protocol was defined over the full

Table 3
Detailed dataset characteristics and patient inclusion summary

Patient ID	Original Seizures	Seizures (Eval)*	Preictal epochs†	Interictal epochs†	Status‡
chb01	7	3	1916	29703	Included
chb02	3	0	0	38457	Excluded
chb03	7	1	672	36071	Included
chb04	4	3	1991	203623	Included
chb05	5	3	1080	30482	Included
chb06	10	5	1621	48575	Included
chb07	3	2	1440	77655	Included
chb08	5	2	1181	5228	Included
chb09	4	3	2160	74856	Included
chb10	7	3	2160	39095	Included
chb11	3	0	0	45567	Excluded
chb12	40	2	312	1958	Included
chb13	12	1	710	27895	Included
chb14	8	3	2114	10893	Included
chb15	20	3	879	8392	Included
chb16	10	1	720	12547	Included
chb17	3	2	1032	15138	Included
chb18	6	1	720	40366	Included
chb19	3	0	0	38373	Excluded
chb20	8	1	437	25949	Included
chb21	4	1	395	35903	Included
chb22	3	3	1825	25123	Included
chb23	7	1	720	20720	Included
chb24	16	0	0	7499	Excluded
Total	198	44	24085	770172	20 incl. / 4 excl.

* Number of independent lead seizures retained after applying postictal and SPH exclusion criteria.

† Epoch counts are reported after removing the 20-minute stabilization period (warm-up).

‡ Patients were excluded if their evaluation segment contained only a single class after warm-up.

cohort, final performance metrics were computed on 20 evaluable patients. Four subjects (chb02, chb11, chb19, and chb24) were excluded from the aggregate analysis because, after applying the temporal labeling rules (30-min preictal window, 5-min SPH, and 4-h postictal exclusion) and the mandatory 20-min warm-up cutoff, the remaining evaluation segments contained only a single class. Specifically, patients chb02, chb11, and chb19 retained no preictal epochs after the exclusion filters, as their original seizures failed to meet the clinical constraints (e.g., occurring within a 4-hour postictal gap of a previous event). In contrast, chb24 was excluded because its only identified preictal block (72 epochs, approx. 3 minutes) occurred entirely within the first 480 epochs of the recording; consequently, this segment was removed by the 20-minute warm-up stabilization required for the temporal features. In such cases, the ROC-AUC is mathematically undefined and the corresponding folds were therefore omitted from the final summary (see Table 3 for a detailed breakdown).

3.5.5. Dimensionality Reduction

Although the 55-dimensional feature space is relatively compact, the potential presence of inter-channel correlations, particularly among spatially adjacent bipolar derivations, motivated an evaluation of dimensionality reduction prior to classification.

Principal Component Analysis (PCA) was assessed as a

candidate preprocessing step. Given the normalised feature matrix $\mathbf{F} \in \mathbb{R}^{N_{epochs} \times 55}$, PCA projects the data onto an orthogonal subspace:

$$\mathbf{z}^{(n)} = \mathbf{W}^T \mathbf{f}^{(n)} \quad (15)$$

where $\mathbf{W}$ contains the eigenvectors associated with the largest eigenvalues of the feature covariance matrix. The number of retained components was selected to explain at least 95% of the total variance.

PCA was ultimately not included in the final pipeline for two reasons. Projecting onto principal components destroys the direct correspondence between individual features and their physiological interpretation, which is necessary for the channel-wise feature importance analysis described in the following section. Moreover, cross-patient validation experiments showed no consistent improvement in AUC when PCA was applied, suggesting that the inter-channel correlations present in the 55-dimensional space do not represent a significant source of noise for tree-based classifiers, which handle correlated features naturally through random subspace sampling.

3.5.6. Performance Metrics

Given the pronounced class imbalance, accuracy alone is not a meaningful performance metric. The following measures were therefore computed for each test subject:

- **Sensitivity:** fraction of preictal epochs correctly identified.
- **Specificity:** fraction of interictal epochs correctly rejected.
- **Precision:** fraction of predicted preictal epochs that are truly preictal.
- **F1-score:** harmonic mean of precision and sensitivity.
- **AUC-ROC:** area under the receiver operating characteristic curve, used as the primary summary metric.

Sensitivity and specificity are defined as:

$$\text{Sensitivity} = \frac{TP}{TP + FN} \quad (16)$$

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (17)$$

where TP , TN , FP , and FN denote true positives, true negatives, false positives, and false negatives, respectively. In addition, the false alarm rate per hour (FAR/h) was computed to provide a clinically grounded assessment of the system's practical applicability. FAR/h was evaluated on the final binary prediction sequence after the warm-up exclusion period and, when enabled, after application of the persistence rule. A false alarm was defined as one contiguous alarm event occurring entirely outside the true preictal class. Consecutive predicted positive epochs were therefore grouped into contiguous alarm clusters, and each cluster containing no true preictal epochs was counted as one false alarm event. The FAR/h was then computed as

$$\text{FAR/h} = \frac{N_{FA}}{T_{\text{interictal}}^{(h)}} \quad (18)$$

where N_{FA} is the number of false alarm events and $T_{\text{interictal}}^{(h)}$ is the total duration of interictal time under evaluation, expressed in hours.

4. Results and Discussion

4.1. Ablation Study and Performance Analysis

To quantify the contribution of each component in the proposed pipeline, an ablation study was performed using four incremental configurations under a LOPO-CV scheme on 20 evaluable subjects from the CHB-MIT Scalp EEG Dataset. Four cases (chb02, chb11, chb19, and chb24) were excluded from ROC aggregation due to the presence of a single class in the retained evaluation segments. The results show the progressive impact of each methodological choice on cross-patient seizure prediction performance.

As reported in Table 5, the Baseline (Raw WPE) configuration achieved a mean AUC-ROC of 55.18%, with low specificity (35.94%), a FAR/h of 0.08 events/hour and a seizure sensitivity of 70.4%. This near-chance performance highlights the limitations of raw WPE as a standalone feature in a subject-independent setting. The primary issue lies in inter-patient variability: preictal entropy may either decrease or increase depending on the subject, resulting in inconsistent predictive patterns that hinder model generalization.

Introducing Patient-Specific Polarity Calibration (WPE+Calibration) led to a substantial performance improvement, increasing AUC-ROC to 67.72% and achieving a sensitivity of 81.69%, with a seizure sensitivity of 81.8%. Aligning entropy directionality across patients therefore appears to be a critical step for enabling cross-subject learning. Once normalized, entropy variations represent deviations from baseline across subjects, allowing the Random Forest classifier to capture meaningful preictal dynamics.

The Full Feature Space configuration, which integrates calibrated WPE with temporal trends, local volatility, and spatial synchrony, yielded the highest discriminative performance with an AUC-ROC of 69.23%, sensitivity 95.58%, F1-score 11.95%, and a pooled seizure sensitivity of 100.0%. These features capture complementary aspects of the preictal transition, enabling the model to represent both the temporal evolution and the spatial coordination of brain dynamics. However, predictions at the single-epoch level remain temporally inconsistent due to noise, resulting in fragmented and unstable decision patterns. This instability manifests as short fluctuations between preictal and interictal states, limiting the clinical usability of raw model outputs.

Thus, to address this limitation, the final configuration (Full Feature Space + Persistence), introduces a temporal consistency constraint, requiring 10 minutes of consecutive preictal predictions ($PERSISTENCE_{EPOCHS} = 240$). This strategy effectively suppresses transient fluctuations, leading to a substantial FAR/h reduction, maintaining 0.25 events/hour, while achieving an AUC-ROC of 69.23%, a sensitivity of 88.33%, and an outstanding pooled seizure sensitivity of 95.5% (per-patient results for this final configuration are reported in Table 4). Specifically, the persistence mechanism operates as a post-processing stabilization mechanism to suppress transient false positives rather than altering the intrinsic discriminative capacity of the model. This trade-off results in more coherent and clinically meaningful alarm patterns.

However, the introduction of a 10-minute persistence rule ($PERSISTENCE_{EPOCHS} = 240$) inherently introduces a deterministic latency in the alarm triggering.

While this constraint is important for suppressing transient false positives and allowing for a clinically acceptable FAR/h, it effectively subtracts 10 minutes from the available SPH. In a real-time deployment, the system would notify the patient only after the persistence condition is met. Future iterations of this framework could investigate adaptive persistence thresholds or weighted voting schemes to minimize this latency without compromising alarm stability.

To further assess whether these incremental improvements are statistically robust rather than attributable to inter-patient variance, paired Wilcoxon signed-rank tests were performed on the per-patient AUC-ROC values, comparing each intermediate configuration (A, B, C) directly against the final configuration D (Table 6). None of the three comparisons reaches conventional statistical significance ($p > 0.05$), although the Baseline vs. D comparison ($p = 0.064$) is the closest to significance and shows by far the largest mean difference (+0.141), consistent with the substantial performance gap already visible in Table 5. Configurations B, C, and D are statistically indistinguishable from one another in terms of AUC-ROC alone ($p = 0.312$ and $p = 0.725$, respectively), given the sample size and inter-patient variability of this cohort ($n = 20$). This is expected for the C vs. D comparison in particular, since the persistence rule operates on the binarized alarm sequence and therefore does not alter the underlying probability ranking from which AUC-ROC is computed. The practical benefit of the additional pipeline components is instead reflected in the event-level metrics reported in Table 5: seizure sensitivity rises from 70.4% (Baseline) to 100.0% (Full Feature Space), and FAR/h drops from 0.54 to 0.25 once persistence is introduced, despite the AUC-ROC remaining statistically unchanged. These results suggest that AUC-ROC alone underestimates the clinical value of the full pipeline, and that event-based metrics are necessary to capture the operational improvements achieved through feature engineering and temporal stabilization.

We attribute this mainly to the small sample size ($n = 20$) combined with the inter-patient heterogeneity documented in subsection 4.5. A subset of patients (the LOW group) stays close to chance regardless of feature configuration. This inflates the variance of the paired differences, so the test lacks power even though the improvement is consistent in direction across nearly all subjects.

4.2. Clinical Utility and False Alarm Reduction

The apparent discrepancy between the low epoch-level specificity (26.81% in the final configuration) and the excellent event-level false alarm rate (0.25 FAR/h) is a deliberate consequence of the pipeline’s design.

The severe class imbalance was explicitly counteracted via cost-sensitive learning and 1:2.5 undersampling, which heavily penalizes missed preictal signatures and shifts the decision boundary to prioritize sensitivity (95.5% at the event level) over epoch-level baseline rejection.

At the same time, the interictal-to-preictal transition is a gradual "complexity collapse," causing some interictal epochs to mimic preictal entropy drops. The model remains intentionally highly sensitive at the micro-scale (5-second epochs), relying on the 10-minute persistence mechanism to act as a robust macro-scale filter. As a result, transient entropy drops do not translate into clinical false alarms,

Table 4

Event-level (seizure-based) evaluation metrics per patient, final configuration (Full Feature Space + Persistence).

Patient ID	AUC	Seizures Pred./Tot.	Seizure sens. (%)	Mean warn. (min)	Median Warn. (min)	FAR/h (events)	FAR/24h (events)	Time in warning (%)
chb01	0.65	3/3	100.0	26.5	31.0	0.194	4.65	80.8
chb03	0.83	1/1	100.0	31.5	31.1	0.838	20.12	68.5
chb04	0.50	3/3	100.0	32.6	35.0	0.163	3.90	95.0
chb05	0.78	3/3	100.0	20.0	18.0	0.189	4.54	42.5
chb06	0.68	4/5	80.0	13.8	7.6	0.267	6.40	39.8
chb07	0.84	2/2	100.0	35.0	35.0	0.093	2.23	89.3
chb08	0.89	2/2	100.0	24.6	24.6	0.275	6.61	73.9
chb09	0.54	3/3	100.0	35.0	35.0	0.212	5.08	84.9
chb10	0.58	3/3	100.0	35.0	35.0	0.368	8.84	82.8
chb12	0.89	1/2	50.0	15.2	15.2	0.000	0.00	91.2
chb13	0.71	1/1	100.0	34.5	34.5	0.568	13.63	59.2
chb14	0.57	3/3	100.0	34.3	35.0	0.000	0.00	97.8
chb15	0.63	3/3	100.0	17.2	13.8	0.000	0.00	97.2
chb16	0.92	1/1	100.0	35.0	35.0	0.230	5.51	70.8
chb17	0.89	2/2	100.0	21.5	21.5	0.476	11.41	40.7
chb18	0.17	1/1	100.0	35.0	35.0	0.178	4.28	94.5
chb20	0.96	1/1	100.0	23.2	23.2	0.000	0.00	99.1
chb21	0.75	1/1	100.0	21.4	21.4	0.201	4.81	59.1
chb22	0.58	3/3	100.0	19.1	22.0	0.573	13.76	40.5
chb23	0.49	1/1	100.0	35.0	35.0	0.208	5.00	47.7
Mean $\pm$ SD	0.69 $\pm$ 0.19	-	96.5 $\pm$ 11.8	27.3 $\pm$ 7.8	27.2 $\pm$ 8.8	0.25 $\pm$ 0.22	6.04 $\pm$ 5.27	72.8 $\pm$ 21.6
Pooled (cohort-wide)	-	42/44	95.5	-	-	-	-	-

Note: aggregate rows (Mean $\pm$ SD, Pooled) summarize the per-patient values above.

Table 5

Ablation study results (mean), including event-level metrics.

Configuration	AUC (%)	Sens. (%)	Spec. (%)	FAR/h	FAR/24h	F1 (%)	Seizure sens. (%)	Median warn. (min)
1. Baseline-Raw WPE	55.18	66.70	35.94	0.089	2.15	7.11	70.4	26.3
2. WPE+Calibration	67.72	81.69	25.92	0.348	8.34	9.41	81.8	26.8
3. Full Feature Space	69.23	95.58	19.82	0.540	12.96	11.95	100.0	28.3
4. Full Feature Space + Persistence	69.23	88.33	26.81	0.252	6.04	11.57	95.5	27.2

Table 6

Paired Wilcoxon signed-rank test: each ablation configuration vs. the final configuration D (n=20 patients).

Comparison	AUC change (D-X)	Wilcoxon test	p-value
A vs D	+0.141	55.0	0.064
B vs D	+0.015	77.0	0.312
C vs D	+0.0002	77.5	0.725

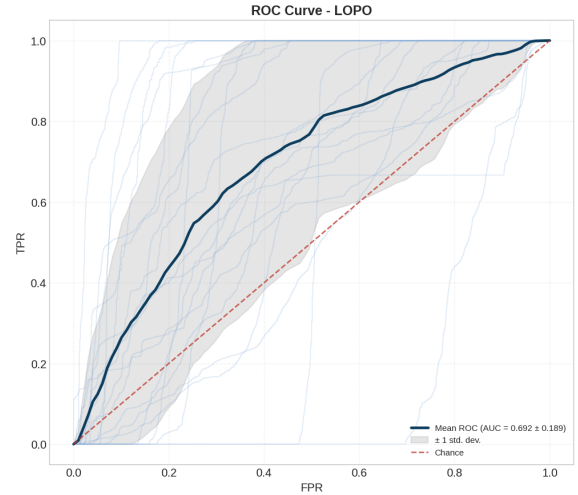
effectively bridging the gap between low epoch-level specificity and high clinical utility.

Minimizing alarm fatigue is critical for practical systems. The final configuration achieves a FAR/h of 0.25 that approximately corresponds to one false alarm every 4 hours. To further quantify restrictiveness, we evaluated the time spent in warning, reflecting the percentage of interictal duration with active alarms.

Although evaluated retrospectively with patient-specific calibration, the persistence mechanism significantly improves temporal stability by suppressing transient output fluctuations. The ROC analysis (Figure 2; mean AUC $\approx$ 0.692) confirms a favorable sensitivity-specificity balance under the conservative LOPO framework.

4.3. Feature Contribution and Explainability

The use of a Random Forest classifier enables interpretability through feature importance analysis based on Mean Decrease in Impurity (MDI). As shown in Figure 3, the top-

**Figure 2:** Mean ROC curve across 20 evaluable patients (LOPO).

ranked features are jointly drawn from the calibrated WPE and local volatility feature families, with the single most informative predictor belonging to the volatility block. This supports the central hypothesis that both the instantaneous level and the short-term dynamics of direction-corrected entropy are informative markers of preictal transitions.

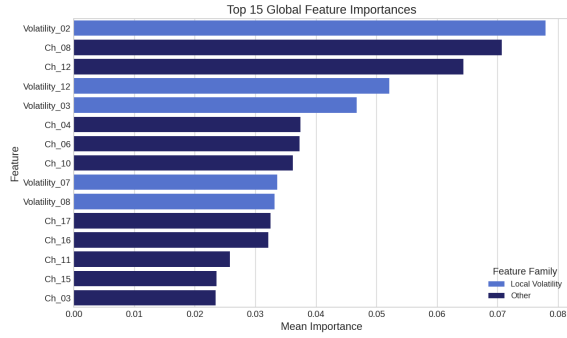


Figure 3: Top 15 predictive features ranked by MDI. The ranking is dominated by features from the calibrated WPE and local volatility families, with only a marginal contribution from spatial synchrony.

4.3.1. Spatial Dominance and Neurophysiological Congruence

As illustrated in Table 7, the feature importance is not uniformly distributed across the channels but shows a clear dominance of posterior (parieto-occipital) regions over anterior (fronto-central) ones. The aggregated contribution of posterior regions reaches 61.85%, nearly double that of anterior regions (36.73%). Within this posterior cluster, the single most informative channel is P4-O2 (11.94%), followed by P3-O1 (10.81%) and T7-P7 (8.20%), together accounting for over 30% of the total predictive power on their own. This spatial pattern aligns with the clinical characteristics of the CHB-MIT dataset, which includes pediatric focal epilepsy originating in temporal and occipital regions. The prominence of channels such as P4-O2 and P3-O1 suggests that the "complexity collapse" is most detectable in the proximity of the epileptogenic zones, where neuronal hypersynchronization begins to override the baseline dynamics prior to seizure onset. In particular, the two leading channels are located on opposite hemispheres (P4-O2 right, P3-O1 left), indicating that posterior involvement is bilateral rather than strictly lateralized to one side.

Table 7

Top 10 EEG channels by feature importance (LOPO mean across 20 patients), aggregated contribution by anterior vs. posterior scalp region, and the least contributive channel.

Channel	Contribution (%)
P4-O2	11.94%
P3-O1	10.81%
F7-T7	10.05%
T7-P7	8.20%
C3-P3	6.38%
F3-C3	6.08%
T8-P8	5.73%
F4-C4	5.70%
P7-O1	5.53%
P8-O2	4.95%
Total posterior regions (P/O channels)	61.85%
Total anterior regions (F/FP channels)	36.73%
FP2-F8 (least contributive)	1.27%

4.3.2. Network-level Dynamics: Role of Synchrony

Global synchrony emerges as a highly informative feature (5.62%), despite being a single scalar among many inputs.

Seizure generation therefore appears not to be a purely local phenomenon, but to involve network-level reorganization.

A reduction in spatial entropy (increased similarity of WPE across channels) serves as a proxy for large-scale cortical coupling, a well-known precursor to seizure propagation.

4.3.3. Temporal Trends and Volatility

The inclusion of temporal trend (δ) and local volatility features (Figure 3) further enhances the model performance, as they capture the dynamic evolution of the preictal state. While the calibrated WPE captures the instantaneous state of the system, these temporal descriptors allow the Random Forest to track the gradual evolution of the preictal transition.

Consistent with previous observations, the contribution of frontal channels remains lower, yet non negligible (e.g., FP1-F7 at 2.37%). Even though the primary seizure focus is located in posterior regions, frontal dynamics still provide complementary information. For instance it may support the model in filtering ocular artifacts or capturing the later stages of preictal propagation.

The feature importance analysis supports a physiologically plausible interpretation of the model, combining local complexity changes with global network dynamics. This argues against the model relying on spurious, black-box correlations.

4.4. Qualitative Analysis of Complexity Collapse

A representative example (Patient chb16, Figure 4) illustrates the practical behavior of the system. The upper panel shows a clear entropy reduction preceding seizure onset, consistent with the hypothesis of a transition toward a more ordered dynamical regime.

The lower panel demonstrates the inverse relationship between WPE and model seizure probability (blue line), with the 10-minute persistence mechanism (orange blocks) effectively filtering transient fluctuations to produce more stable alarm intervals. This qualitative illustration shows how sustained preictal signatures can generate visible lead time before seizure onset, although single-patient examples should be interpreted as demonstrative rather than quantitative evidence of clinical performance.

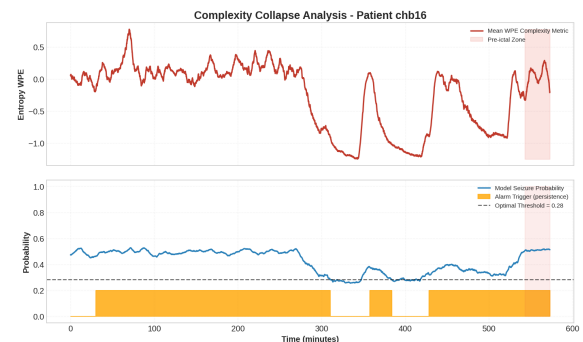


Figure 4: Qualitative analysis (Patient chb16). The probability (blue) rises as entropy drops, triggering a persistent alarm (orange).

4.5. Inter-Patient Heterogeneity Analysis

To assess the impact of inter-subject variability, patients were divided into two groups based on the median AUC-ROC (0.698) as the splitting criterion.

The HIGH group ($n = 10$) includes subjects with the highest predictability: *chb03*, *chb05*, *chb07*, *chb08*, *chb12*, *chb13*, *chb16*, *chb17*, *chb20*, and *chb21*. Conversely, the LOW group ($n = 10$) comprises subjects where preictal discriminability was lower: *chb01*, *chb04*, *chb06*, *chb09*, *chb10*, *chb14*, *chb15*, *chb18*, *chb22*, and *chb23*.

An intriguing observation arises from the longitudinal analysis of Subject 01. Case *chb01* (recorded earlier) falls in the LOW group, while *chb21* (from the same patient 1.5 years later) belongs to the HIGH group. This points to a temporal evolution in the detectability of the WPE signature within the same individual.

4.5.1. Performance Stratification (LOPO Analysis)

To characterize subgroup-specific performance, we conducted LOPO cross-validation within each subgroup. As reported in Table 8, substantial differences emerged: the HIGH group achieved a mean AUC-ROC of 84.61%, indicating that WPE captures robust preictal dynamics for this subset of patients.

Conversely, the LOW group showed near-chance performance (AUC 53.84%). The WPE distribution in Figure 5 reveals greater overlap between interictal and preictal states for these patients, suggesting either subtler dynamical transitions or characteristics that reduce the discriminability of the complexity collapse signature.

Table 8

LOPO Performance comparison between HIGH and LOW subgroups.

Group	AUC (%)	Sens. (%)	Spec. (%)	FAR/h	F1 (%)
HIGH	84.61%	92.5 %	29.5%	0.29	12.1%
LOW	53.84%	84.1%	24.1%	0.22	11.00%

4.5.2. Asymmetric Generalization and Regularization Potential

To evaluate the intrinsic robustness of the learned features, we conducted a cross-group generalization stress test, logically distinct from the LOPO validation. In this scenario, the Random Forest was trained on one entire subgroup and tested on the other. The results, summarized in Table 9, indicate that the learned representation transfers reasonably well in both directions, without the sharp asymmetry that might be expected between a "high-performing" and a "low-performing" subgroup. A model trained exclusively on the LOW group achieved an AUC-ROC of 77.8% when tested on the HIGH group, a moderate decrease of 6.8 percentage points relative to the HIGH group's own within-group LOPO baseline (84.61%, Table 8). Conversely, a model trained on the HIGH group and tested on the LOW group achieved an AUC-ROC of 60.2% — not a degradation, but an increase of 6.4 percentage points relative to the LOW group's own within-group baseline (53.84%, Table 9).

This asymmetry is therefore the opposite of a simple failure-to-generalize pattern: the LOW group, which shows the weakest within-group discriminability, benefits from being evaluated with a classifier trained on the more homogeneous and reliable preictal signatures of the HIGH group,

while the HIGH group experiences only a modest performance cost when the classifier is trained on the noisier LOW-group data. This result suggests that the two subgroups encode partially shared, transferable preictal dynamics rather than group-specific artefacts, and that exposure to a broader range of patient profiles during training does not compromise, and may even support, generalization to more challenging cases. Event-level metrics support this interpretation: seizure sensitivity remains high in both directions (93.8% for LOW→HIGH and 82.1% for HIGH→LOW, Table 9), with comparable false alarm rates (FAR/h = 0.27 in both configurations).

Table 9

Cross-group generalization stress test, including event-level metrics.

Training Set	Test Set	AUC (%)	Δ Baseline	FAR/h	Seizure Sens. (%)
LOW	HIGH	77.8%	-6.8%	0.27	93.8
HIGH	LOW	60.2%	+6.4%	0.27	82.1

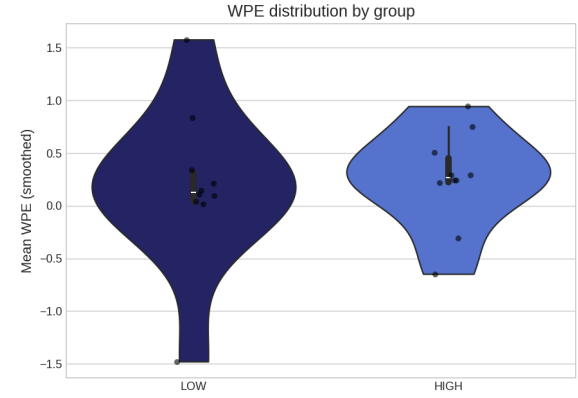


Figure 5: Distribution of WPE across subgroups. The overlap in the LOW group highlights the challenge of identifying a universal complexity threshold.

4.6. Comparison with State of the Art

To contextualize the performance of the proposed framework, a comparison was made with recent seizure prediction studies evaluated on the CHB-MIT Scalp EEG Database. Particular attention was given to cross-patient validation protocols, such as Leave-One-Patient-Out (LOPO) and Leave-One-Out (LOO), since the evaluation strategy strongly affects the reported performance and the extent to which patient-specific information can be exploited. Studies based on patient-specific or randomly partitioned validation schemes were therefore not considered directly comparable with the cross-patient setting adopted in this work.

Under the adopted evaluation protocol, the proposed framework achieves a mean AUROC of 0.692, with a sensitivity of 88.3% and, per-patient, a median false alarm rate of 0.20 events/h (vs. the cohort mean of 0.25 events/h reported in Table 5; the difference reflects the right-skewed distribution of FAR/h across patients visible in Table 4, e.g. *chb03* at 0.84 events/h). These results indicate that the model is able to identify a substantial proportion of preictal periods while maintaining a relatively low false-alarm burden. The AUROC, however, is lower than that reported by some of the most recent deep-learning approaches, suggesting that the main strength of the proposed framework does not lie

Table 10

Comparison with recent cross-patient seizure prediction studies on the CHB-MIT dataset.

Study	Method	Evaluation protocol	Target-patient adaptation	AUROC	Sensitivity	FAR/h
This work	WPE + Random Forest + temporal/spatial features + persistence	LOPO, 20 evaluable patients	Polarity calibration + Robust baseline scaling; no RF retraining	0.69	88.3%	0.2
Jemal et al. [25]	CNN, cross-subject baseline	LOPO, CHB-MIT (22 patients)	None	0.69	N/R	N/R
Jemal et al. [25]	CNN + CDAN+E	LOPO, CHB-MIT (22 patients)	Unsupervised domain adaptation	0.75	N/R	N/R
Shafieezadeh et al. [26]	XGBoost + Cal1	LOO, CHB-MIT (8 patients)	Fine-tuning using 1 target seizure	N/R	$52.89 \pm 11.66\%$	N/R
Shafieezadeh et al. [26]	XGBoost + Cal2	LOO, CHB-MIT (8 patients)	Fine-tuning using 2 target seizures	N/R	$58.44 \pm 10.47\%$	N/R
Chen et al. [27] ^a	CG-MambaNet (CNN-GCN-Mamba-BiLSTM)	Strict LOPO $\times$ 5 seeds, CHB-MIT (22 patients)	None	0.8152 ± 0.0176	N/R	0.32^b

^a 2026 preprint; not peer-reviewed at the time of writing.^b Event-level false alarms/hour after temporal persistence filtering; therefore, this value may not be directly equivalent to the FAR/h definition used in the present study.

N/R: not reported. LOPO: Leave-One-Patient-Out. LOO: Leave-One-Out. RF: Random Forest.

in maximising overall score-based discrimination. Rather, its performance reflects a favourable operating regime characterised by high sensitivity and limited false alarms.

A relevant comparison is provided by Jemal et al. [25], who evaluated seizure prediction on CHB-MIT using a cross-subject LOPO protocol. Their conventional CNN baseline achieved an AUROC of 0.69, while the introduction of a Conditional Domain Adversarial Network with entropy conditioning (CDAN+E) increased the AUROC to 0.75. The AUROC of 0.69 obtained by the proposed framework is therefore comparable to the conventional cross-subject baseline and remains below the domain-adaptation model. This comparison highlights the difficulty of achieving robust generalisation across patients and suggests that explicitly addressing inter-subject distribution shifts can provide additional benefits.

The proposed framework addresses inter-patient variability through a different mechanism. Instead of adapting the classifier through domain-adversarial training, it models the patient-dependent behaviour of WPE and performs subject-level calibration before classification. In particular, the direction of the preictal entropy variation is determined for each patient and used to align the WPE representation across subjects. This approach is motivated by the substantial heterogeneity observed in the entropy dynamics, where the preictal transition does not necessarily correspond to the same direction of change for all patients. The ablation analysis previously reported in this work supports the importance of this calibration step, showing that the predictive value of raw WPE substantially improves once inter-subject differences in entropy polarity are explicitly taken into account.

A conceptually related approach is presented by Shafieezadeh et al. [26], who investigate patient-independent seizure forecasting with target-patient calibration. Their XGBoost baseline evaluated under LOO achieves a sensitivity of $44.02 \pm 10.38\%$, which increases to $52.89 \pm 11.66\%$

when one seizure from the target patient is used for calibration and to $58.44 \pm 10.47\%$ when two seizures are used. Although the evaluation metrics are not directly equivalent to those reported in the present study, these results support the broader observation that incorporating patient-specific information can improve seizure forecasting in cross-patient settings.

The nature of the calibration is nevertheless different between the two approaches. Shafieezadeh et al. perform target-patient fine-tuning using labelled seizures, thereby adapting the predictive model itself. In contrast, the proposed framework does not retrain the Random Forest using target-patient seizures. Patient-specific information is instead used at the representation level to estimate the polarity of the entropy variation and the corresponding normalisation parameters, after which the same shared classifier is applied. The proposed strategy can therefore be regarded as representation-level calibration rather than target-patient classifier retraining. This distinction reduces the amount of model adaptation required at deployment, while still allowing the representation to account for patient-specific characteristics.

The most demanding comparison is provided by the recent CG-MambaNet architecture [27]. This model combines convolutional, graph-based, Mamba and bidirectional recurrent components and reports an AUROC of 0.8152 ± 0.0176 under strict LOPO evaluation on CHB-MIT. Its reported event-level false alarm rate after temporal persistence filtering is 0.32 alarms/h. In terms of AUROC, CG-MambaNet therefore clearly outperforms the proposed framework.

However, this comparison also illustrates a substantial difference in modelling complexity. CG-MambaNet learns a high-dimensional spatio-temporal representation through a multi-component deep architecture, whereas the proposed framework relies on an explicitly engineered feature representation based on WPE and its temporal and spatial dynamics, followed by a compact Random Forest classifier. The

lower AUROC of the proposed approach should therefore be interpreted together with its substantially simpler and more transparent modelling strategy. The model does not aim to reproduce the representational capacity of large end-to-end deep architectures, but rather to investigate whether an interpretable entropy-based biomarker can retain useful predictive information across heterogeneous patients.

The false alarm results provide an additional perspective on this comparison. The proposed framework achieves a median FAR/h of 0.20, indicating that the typical patient-level false-alarm burden remains relatively low despite the high sensitivity of 88.3%. This is particularly relevant because AUROC alone does not fully characterise the practical behaviour of a seizure prediction system. In a real-time forecasting scenario, the ability to maintain a high detection rate while limiting the number of false alarms is an important aspect of system usability. Nevertheless, the FAR/h values should be compared cautiously across studies because false alarms may be defined and aggregated differently, and the CG-MambaNet value is explicitly reported as an event-level rate after temporal persistence filtering.

Overall, the comparison indicates that the proposed framework does not achieve the highest absolute AUROC among the recent cross-patient methods considered. Its contribution instead lies in demonstrating that an entropy-based representation, combined with subject-level calibration and a relatively lightweight classifier, can provide meaningful cross-patient seizure prediction while maintaining a high sensitivity and a low median false-alarm rate. This represents a different trade-off from increasingly complex deep-learning architectures: rather than maximising discriminative performance at the expense of model complexity, the proposed approach prioritises interpretability, compactness, and an operationally favourable sensitivity–false-alarm balance.

The reported performance should also be interpreted in light of the calibration procedure adopted in this study. The framework is not a fully target-blind, zero-calibration predictor, since patient-specific information is required to estimate the entropy polarity and normalisation parameters. This is the same enrollment-based calibration described in subsection 3.4.1, and it should be kept in mind both when comparing our numbers with fully target-blind methods and when considering real-world deployment: the system requires a brief per-patient labeling step, it is not a zero-shot predictor.

5. Conclusions

This work introduces a novel perspective on seizure prediction by framing the preictal transition as a shift in signal complexity and by explicitly addressing inter-patient variability through a polarity calibration mechanism. Departing from traditional energy-based approaches, we modeled the preictal transition as a change in the organization of brain dynamics, hypothesizing that seizure onset is preceded by a measurable shift in spectral complexity.

To capture this phenomenon, we employed WPE as a dynamic biomarker and designed a feature representation that integrates instantaneous entropy, temporal evolution, local variability, and spatial synchrony. A key contribution of this study is the introduction of a patient-specific polarity calibration mechanism, which addresses the previously overlooked issue of inter-subject variability in the direction

of entropy change. By aligning entropy dynamics across patients, this step enables effective cross-subject learning while preserving interpretability.

The proposed framework was evaluated on the CHB-MIT dataset under a Leave-One-Patient-Out protocol with subject-level calibration. The results indicate that calibrated entropy features retain predictive value across heterogeneous patients, achieving a mean AUC-ROC of 0.692, with a pooled seizure-level sensitivity of 0.95 (42/44 seizures correctly predicted) and a mean false alarm rate of 0.25 per hour. Moreover, the use of a Random Forest classifier allows for feature-level interpretability, revealing physiologically plausible spatial and temporal patterns associated with seizure generation. These findings support the idea that entropy, when properly modeled and calibrated, can serve as a meaningful and clinically relevant biomarker for seizure prediction.

6. Limitations and future works

These results should nonetheless be read against several constraints of the current design. The most consequential concerns the subject-specific calibration step, which relies on access to labeled data from each patient. While this assumption is consistent with realistic clinical workflows, it prevents the system from operating in a fully blind, zero-calibration setting. Specifically, polarity calibration is estimated from each patient’s complete recording, including all seizure events, which implicitly assumes this information would be available from an initial calibration phase; unlike the baseline normalization step (median/IQR scaling from the first 20 minutes of interictal recording), this calibration is not strictly causal within the current cross-validation scheme. A majority-vote alternative using only training-patient directions was tested but yielded a drop in mean AUC-ROC (0.69→0.57), confirming that WPE polarity is patient-specific rather than shared across the cohort; a fully prospective calibration (e.g., derived from a patient’s first seizure only) is left for future work.

A separate concern is that the evaluation protocol includes retrospective elements, such as the selection of operating thresholds and the use of a persistence rule, which may not directly translate to real-time deployment scenarios. As a result, the reported performance should be interpreted as an upper-bound estimate under controlled conditions.

Another important limitation concerns inter-patient variability. The analysis revealed substantial differences in performance across subjects, with a subset of patients exhibiting near-chance predictability. Entropy-based features, while informative, therefore appear not to fully capture all possible preictal patterns, reflecting the highly heterogeneous nature of seizure generation mechanisms. A related caveat concerns the independence of the LOPO folds: cases chb01 and chb21 correspond to the same patient, recorded 1.5 years apart. Although these two recordings are treated as independent test folds and do not share any epochs, they may still exhibit correlated underlying physiology, meaning that the cross-validation folds are not strictly independent in the statistical sense. This could result in a mild, patient-specific inflation of the estimated cross-validation performance for this subject, and should be considered when interpreting the reported per-patient results.

Additionally, the current framework relies exclusively on EEG data, potentially overlooking complementary physio-

logical signals that could improve predictive performance.

A further limitation is that the CHB-MIT dataset consists exclusively of pediatric patients (ages 1.5-22 years) with drug-resistant epilepsy. Since brain dynamics, EEG spectral characteristics, and the physiological substrates of neuronal hypersynchronization differ substantially between the pediatric and adult brain, the generalizability of the proposed WPE-based biomarker and the specific polarity calibration behavior to adult epilepsy populations remains untested and should not be assumed. Validation on adult EEG datasets is required before any claims of broader clinical applicability can be made.

A further limitation concerns the HIGH/LOW subgroup analysis (subsection 4.5): with only 20 evaluable patients split into two groups of $n = 10$, the AUC-ROC estimates reported in Tables 8 and 9 carry substantial statistical uncertainty. While the observed patterns are consistent with the broader trends discussed elsewhere in the paper, they should be regarded as exploratory and hypothesis-generating rather than as robust, confirmed subgroup effects. Validation on a larger cohort would be needed before drawing firm conclusions about differential predictability across patient profiles.

Future work will focus on addressing these limitations and extending the proposed approach toward more robust and clinically applicable systems. One promising direction is the integration of multimodal physiological signals, such as heart rate variability or other autonomic indicators, to provide a more comprehensive characterization of the preictal state. Another avenue involves the development of adaptive or online learning strategies, allowing the model to continuously update its parameters in response to evolving patient-specific dynamics. Ultimately, implementing the pipeline on lightweight, edge-based hardware represents a crucial step toward real-world deployment, with the long-term goal of enabling wearable and non-invasive seizure prediction systems.

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